

Question 1:

The comparison of the energy per day produced by identical solar fields was performed based on the geographical location of Toronto, as well as Bogota. Toronto, being a Canadian city, is considerably farther away from the equator than Bogota. Additionally, the Sun is more directly overhead (higher altitude angle) in Bogota throughout the year than in Bogota. This can be observed in Table 1.

Date	Toronto	Bogota
March 21st	46	85
June 21st	70	71
September 21st	47	86
December 21st	24	63

Table 1: Altitude angles in Toronto and Bogota throughout the year

The effect of this change in altitude angle can be observed by the optimum angles of the heliostats for each city, as seen in Figure 1 and 2. In Bogota, the Sun is more directly overhead throughout the year, and the tilt angle of the panels is smaller than in Toronto on average.

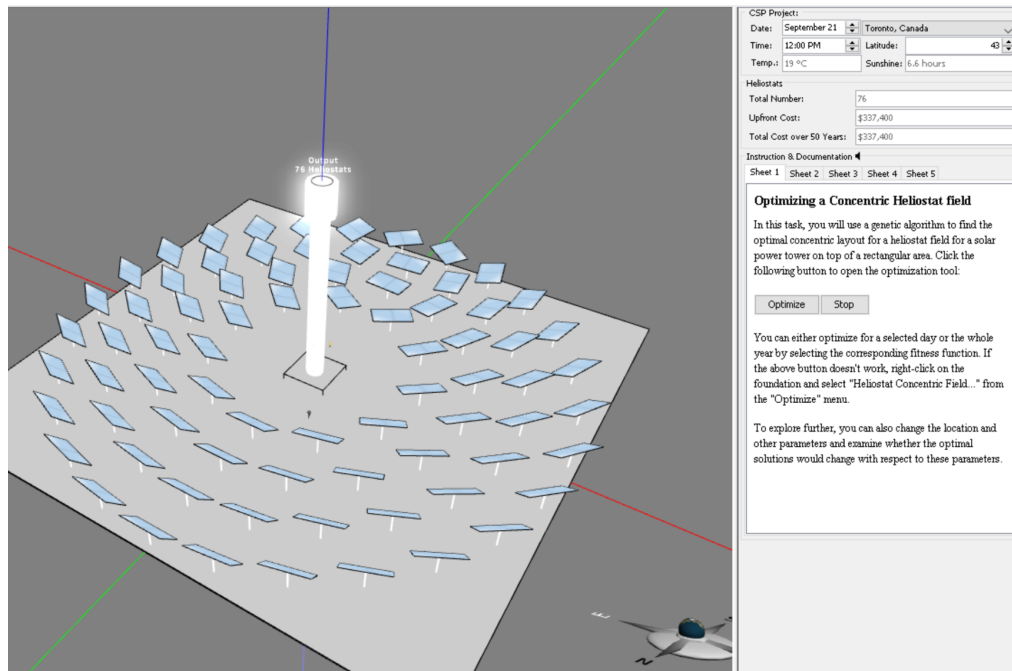


Figure 1: Solar field at optimum angle in Toronto on September 21st at noon

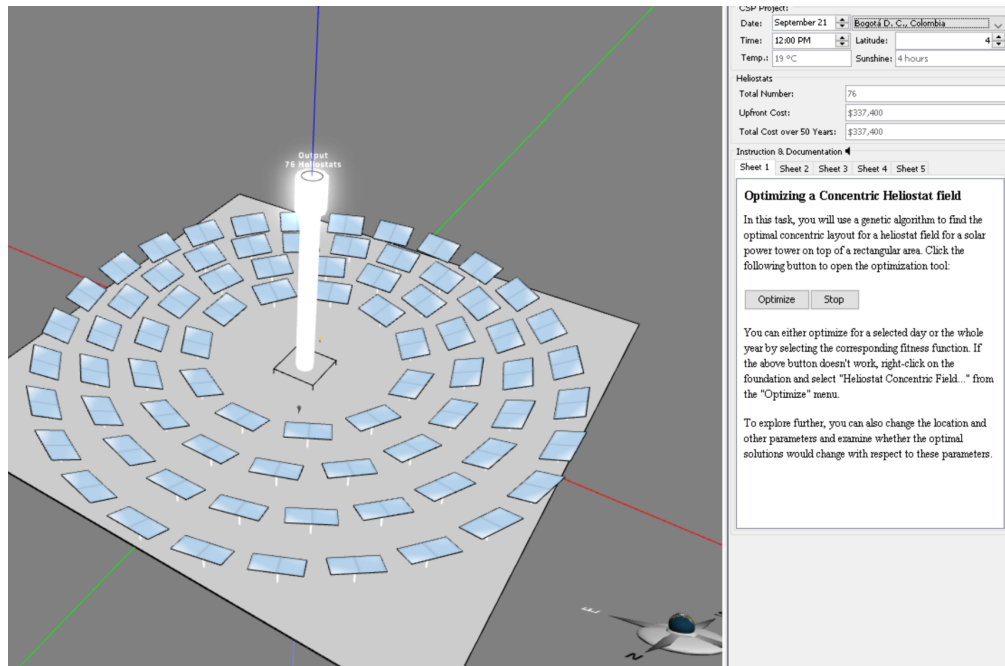


Figure 2: Solar field at optimum angle in Bogota on September 21st at noon

When the simulation was run, it was observed that the total annual electricity generated in Toronto was 235020 kWh (average 643.89kWh per day), and the Energy per Day graph may be seen in Figure 3.

In Bogota, the same simulation was run under identical conditions and the total annual electricity generated was 265969 kWh (average 728.68kWh per day). Figure 4 depicts the Energy per Day in Bogota throughout the year.

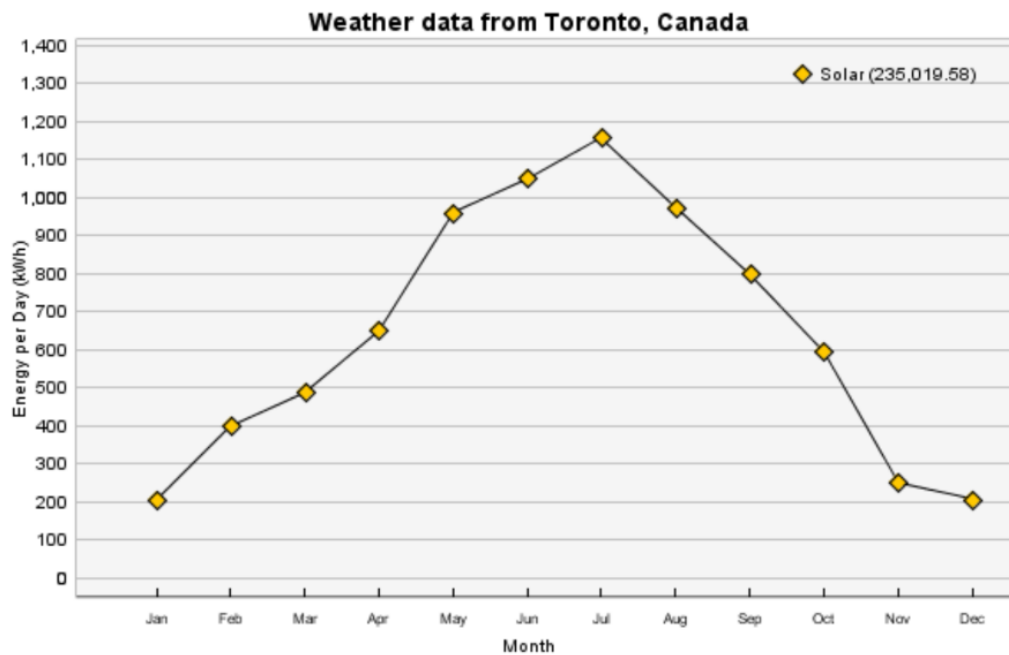


Figure 3: Graph depicting the generated energy per day in Toronto, Canada

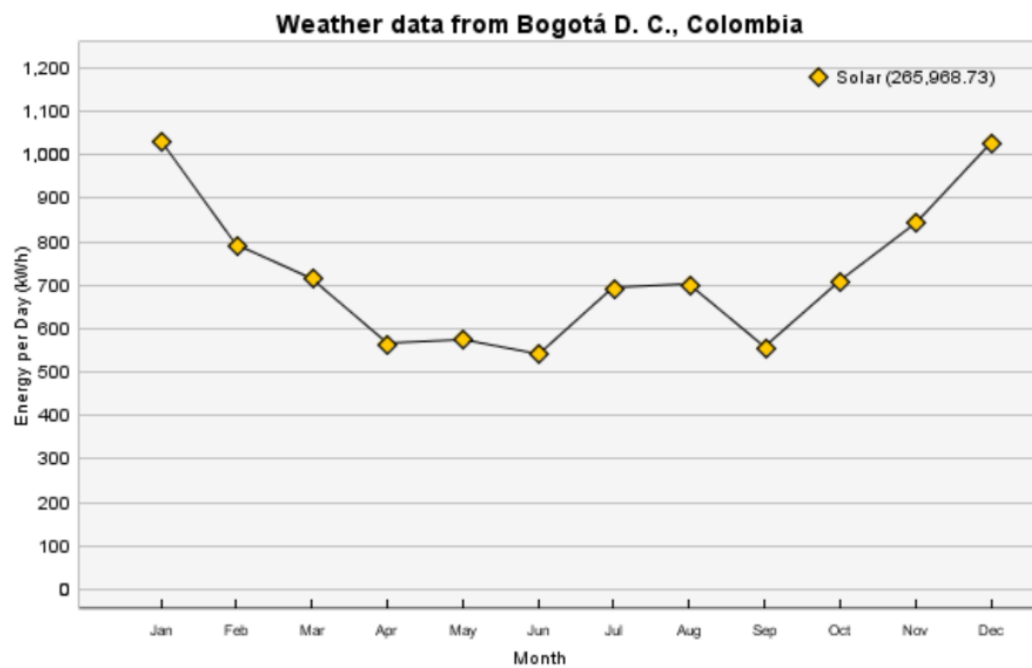


Figure 4: Graph depicting the generated energy per day in Bogota, Colombia

The average energy generated per day was greater in Bogota than in Toronto, which may be due to the greater extent of direct and high-altitude angle sunlight that is received in Bogota than in Toronto.

Question 2:

There were 8 different cases of varying heliostat sizes provided. These size parameters were used to run 8 simulations on Energy3D and output the respective yearly electricity generated. Based on a cost analysis over 5 years, with an installation cost of \$40/m² and an electricity price of \$0.20/kWh, it was found that case 1 was the most profitable (\$209,859.32), as seen in Table 2.

Case	Width (m)	Height (m)	Yearly Electricity Generated (kWh)	Electricity Revenue Over 5 Years	Installation Cost	Net Profit
1	6	6	319,299.32	\$319,299.32	\$109,440.00	\$209,859.32
2	6	4	238,287.45	\$238,287.45	\$72,960.00	\$165,327.45
3	4	4	161,976.58	\$161,976.58	\$48,640.00	\$113,336.58
4	4	6	226,903.99	\$226,903.99	\$72,960.00	\$153,943.99
5	5	3	153,609.36	\$153,609.36	\$45,600.00	\$108,009.36
6	3	5	150,635.78	\$150,635.78	\$45,600.00	\$105,035.78
7	5	5	242,960.02	\$242,960.02	\$76,000.00	\$166,960.02
8	3	3	92,918.77	\$92,918.77	\$27,360.00	\$65,558.77

Table 2: Net profits of 8 pre-defined solar field cases

This case had a heliostat size of 6m x 6m, which was the largest size available among the 8 cases. Under the given installation cost and price of electricity, the results indicate the efficiency of a heliostat is proportional to the size of the heliostat.

Question 3:

For this question, the variable parameter was the height of the solar tower that is absorbing the incoming light from the surrounding heliostats. As the height of the solar tower increased, the required tilt angle of the heliostats decreased, as can be seen by Figures 5, 6, and 7.

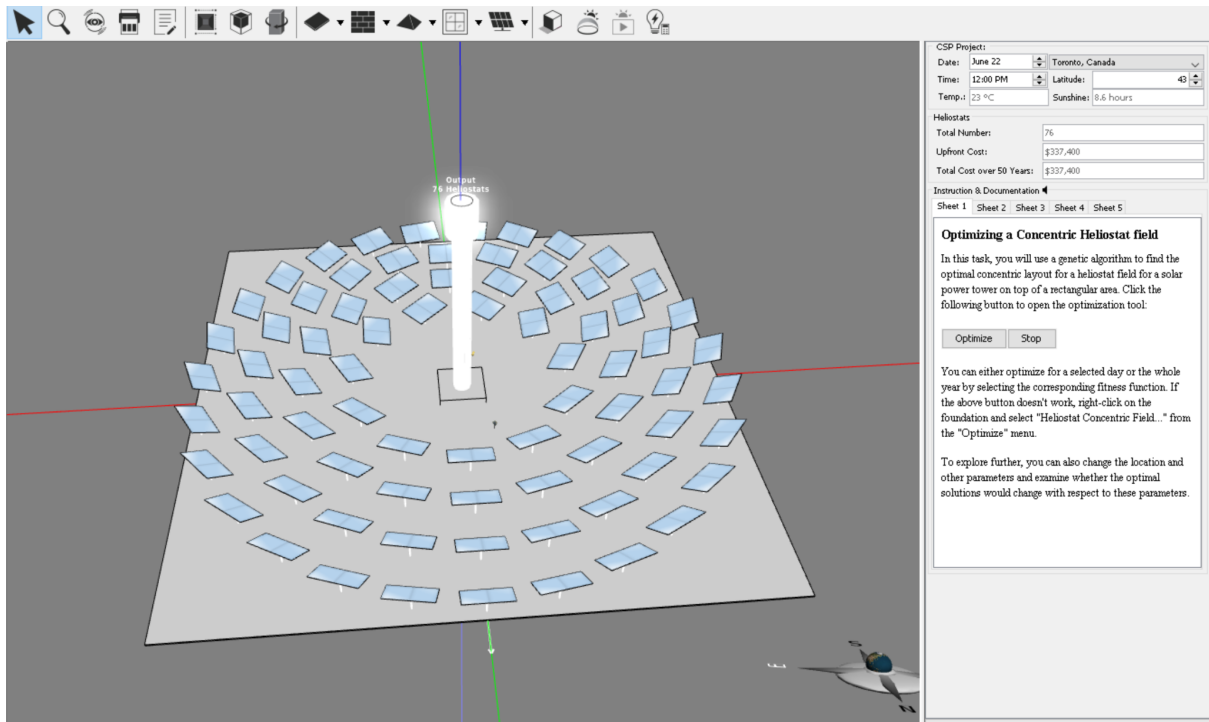


Figure 5: Solar field with solar tower of height 30m

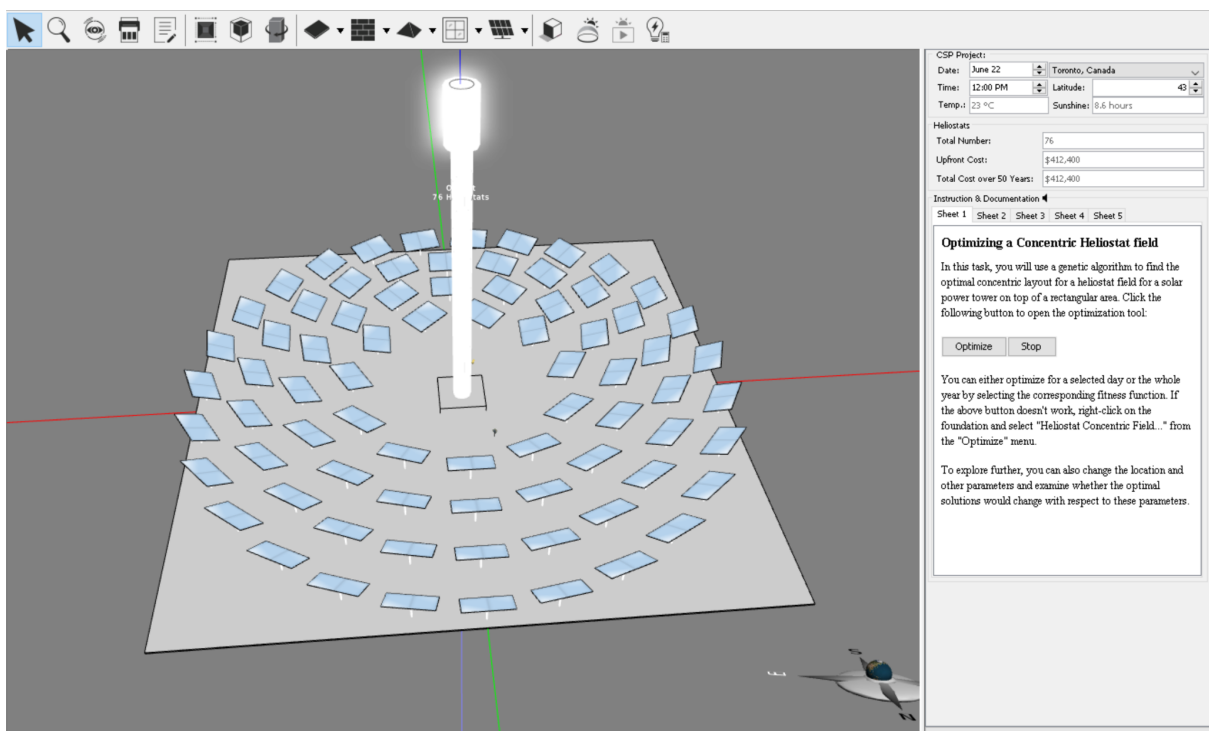


Figure 6: Solar field with solar tower of height 45m

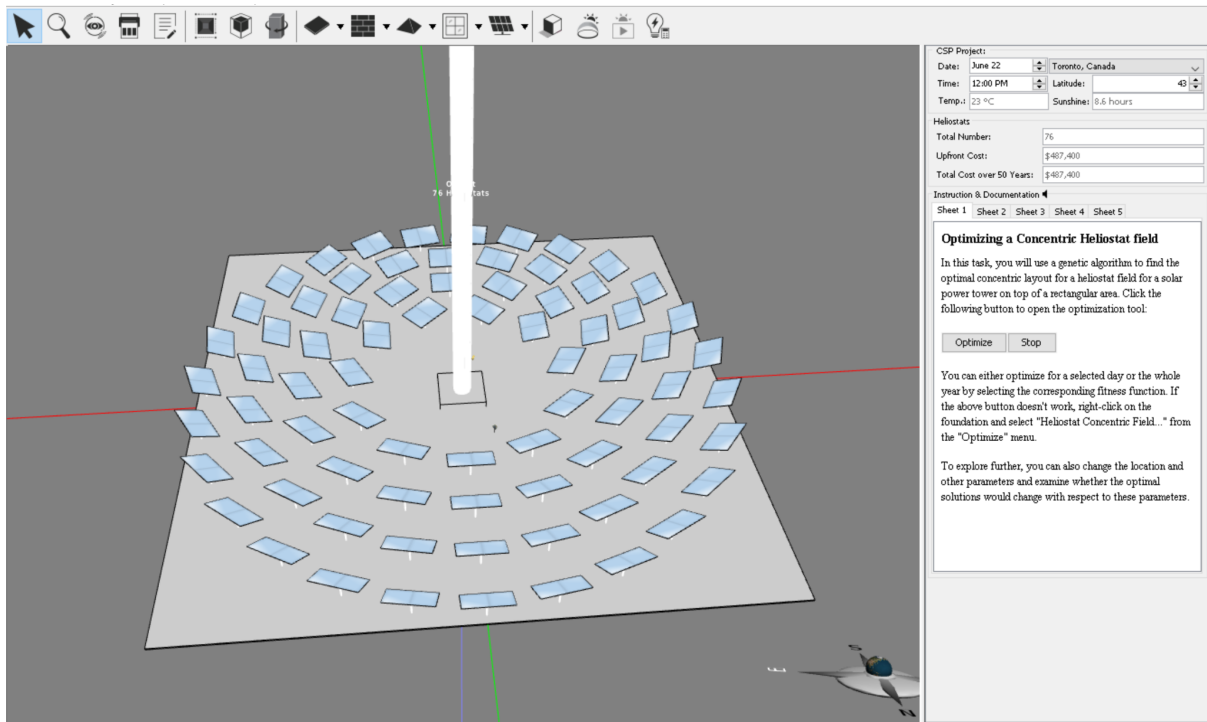


Figure 7: Solar field with solar tower of height 60m

At these 3 different configurations, the simulation was run to determine the annual electricity generated. The results can be seen by the Figures 8, 9, and 10 in terms of expected energy generated per day.

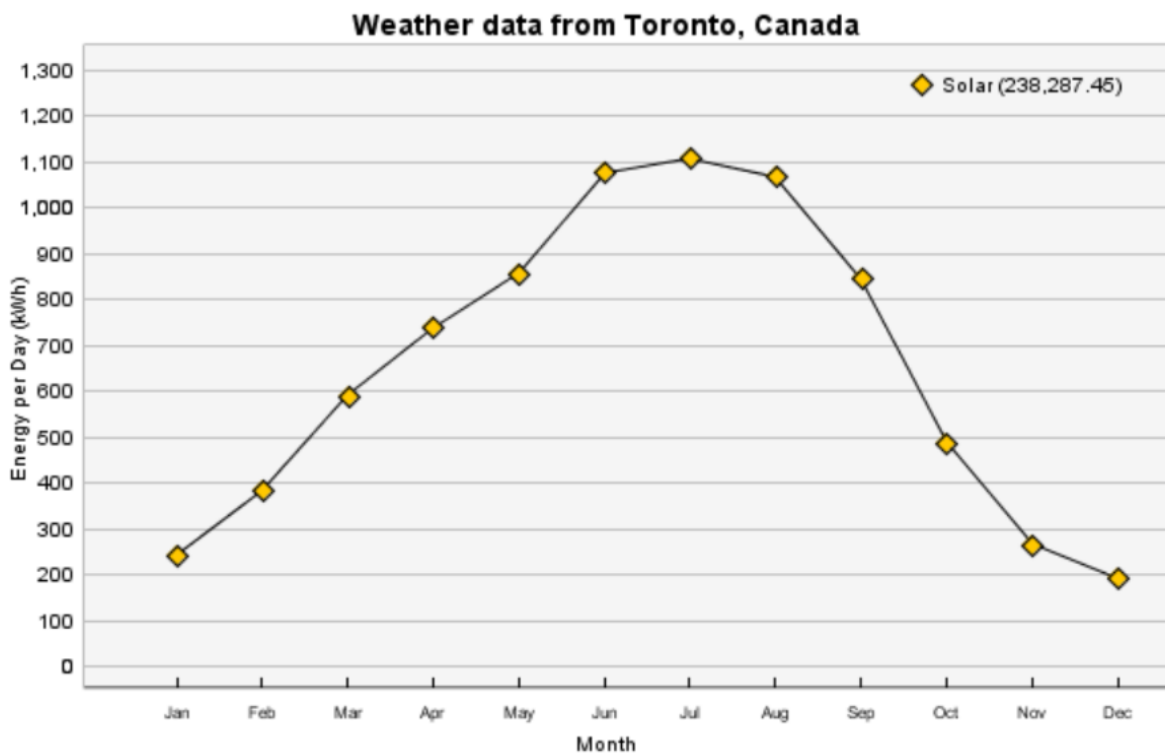


Figure 8: Graph depicting the generated energy per day at solar height = 30m

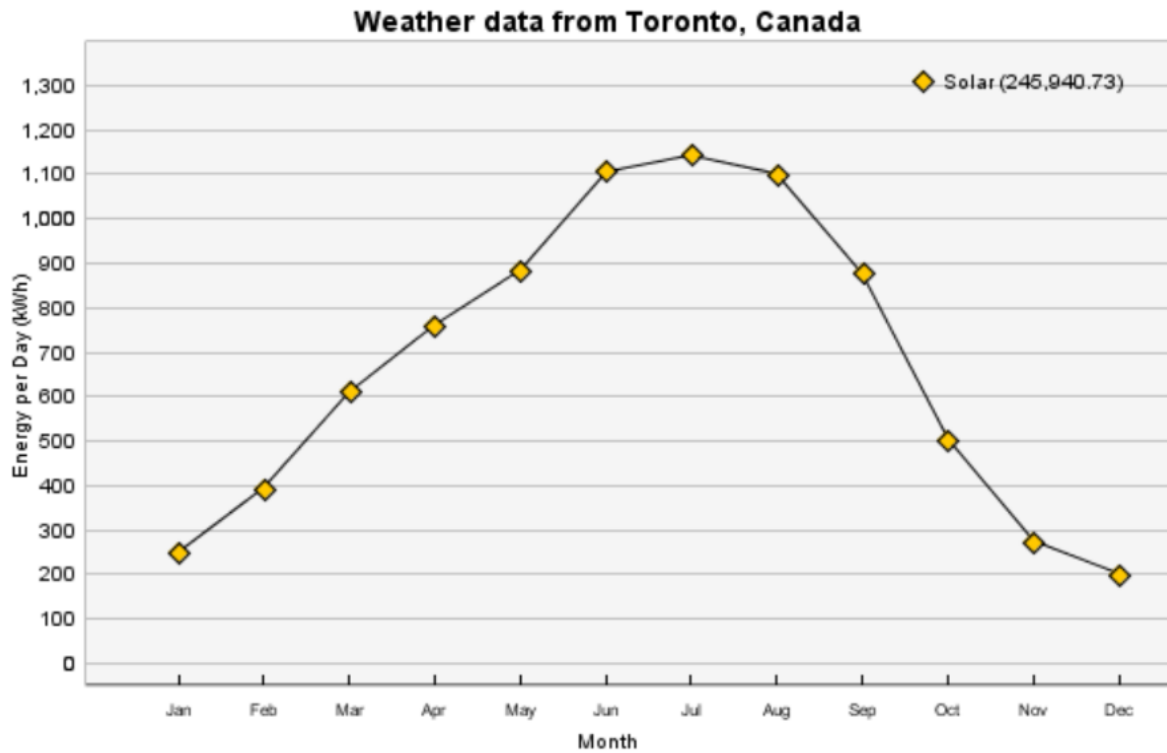


Figure 9: Graph depicting the generated energy per day at solar height = 45m

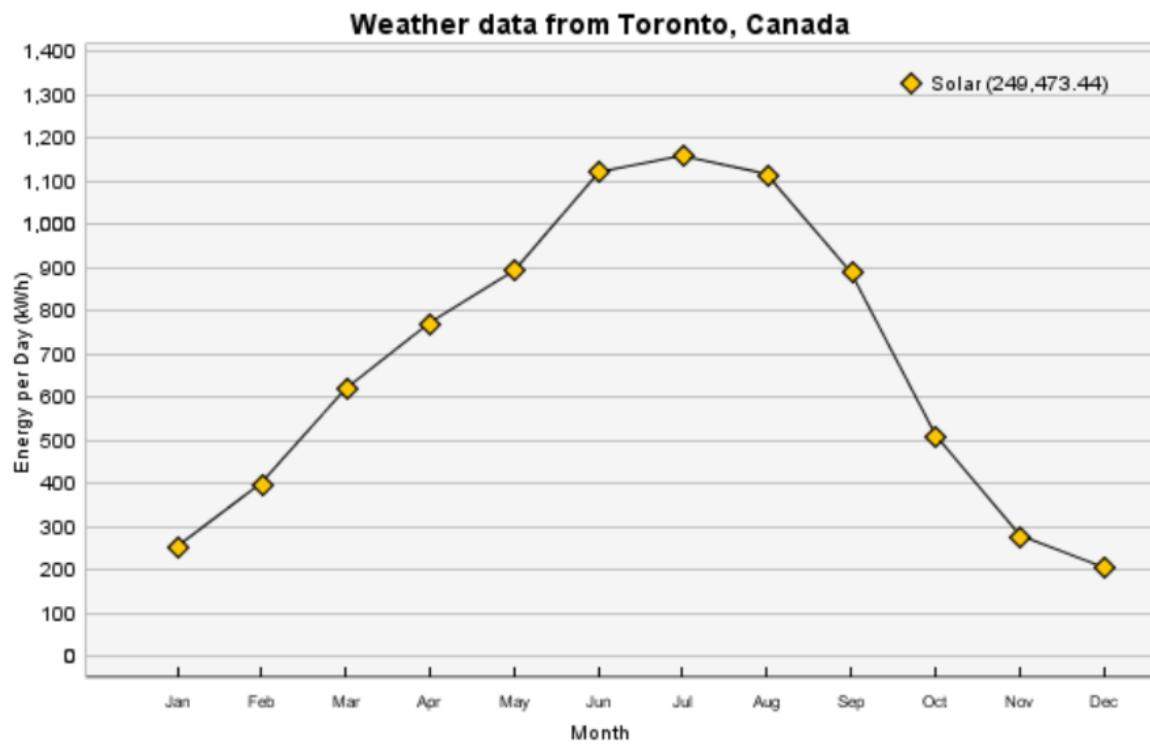


Figure 10: Graph depicting the generated energy per day at solar height = 60m

The results from these simulations indicate that the highest overall average daily electricity generated (683.49kWh) occurred when the height of the solar tower was the highest, as seen in Table 3. This is because the lower required tilt angle of the heliostats allow them to more effectively use their total projected surface area to reflect light towards the solar tower. This indicates that the electricity generated is proportional to the height of the solar tower.

Height of Solar Tower (m)	Yearly Electricity Generated (kWh)	Average Daily Electricity Generated (kWh)
30	238287.45	652.84
45	245940.73	673.81
60	249473.44	683.49

Table 3: Results of Electricity Generated at varying solar tower heights